9 Polarization

9.1 Transverse and longitudinal waves

We have seen that an elastic wave in a string is **transverse**, meaning that the perturbation is in the direction transverse to the direction of propagation of the wave. On the other hand, a sound wave in a fluid is **longitudinal**, meaning that the perturbation is also in the direction that the wave is travelling in. In the first case, we know this because we can actually see the wave travelling along the string. And in the second case, we infer that the wave is longitudinal, because of properties of the medium.

But what about an electromagnetic wave, travelling in free space? Is it a transverse or a longitudinal wave? How can we tell?

9.2 Electromagnetic waves are transverse

Today, we have a very simple way of proving that light is in fact a transverse wave, thanks to an invention by Edwin Land in 1932, when he was just 19 years old: The **polaroid**, or **polarizing filter**. A sheet of this material placed in front of a computer display will let light through to varying degrees, and even extinguish it altogether, simply by rotating it. A longitudinal wave is completely symmetrical about the direction of propagation (think of a slinky; if the slinky is rotated about its axis, the wave looks exactly the same). Therefore, light must be a transverse wave.

James Clerk Maxwell had already proved this mathematically, by showing that the solutions to the wave equation derived from his equations for the electromagnetic field were indeed transverse waves. From Maxwell's theory, we get the following picture of an electromagnetic wave: There is an oscillating electric field (vector), and an oscillating magnetic field (vector) at right angles to each other, and both transverse to the direction of propagation of the wave.

To make it easier to represent the wave, we can drop the magnetic field, and just draw the electric field vector. If we let the wave propagate in the z direction, the electric field vector is contained in the (x, y) plane,

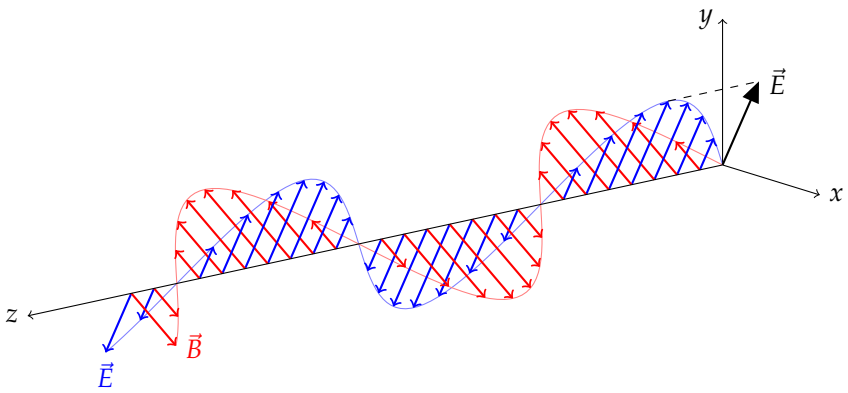


Figure 9.1: Snapshot of an electromagnetic wave, consisting of oscillating electric and magnetic fields at right angles to each other, and also perpendicular to the direction of propagation of the wave, which we choose to be the z direction. At any point on the z axis $E_z = 0$, so $\vec{E}$ is in the (x, y) plane.

so we can draw it in 2D and still retain all the pertinent information about the wave. An example is shown in Figure 9.2.

In the example of Figures 9.1 and 9.2, $\vec{E}$ oscillates in the same direction all the time. The wave is said to be **linearly polarized**. We can decompose $\vec{E}$ into x and y components, each one of which is a wave propagating in the z direction in its own right:

$$\tilde{E}_x = E_{0x} e^{i(\omega t - kz)} \tag{9.1a}$$

$$\tilde{E}_y = E_{0y} e^{i(\omega t - kz)} \tag{9.1b}$$

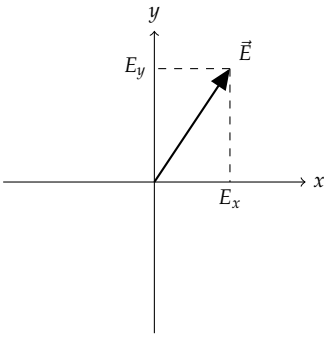


Figure 9.2: Just the electric field vector. Bear in mind that $\vec{E}$ is a function of time, so this is just a snapshot.

Notice that each component is represented by a phasor.

9.3 *The effect of a polarizing filter on a linearly polarized wave*

Imagine an electromagnetic wave reaching a thin conducting wire. The electrons in the wire are accelerated along the wire by the component of the electric field parallel to the wire, and lose energy to the wire. But the electrons cannot be accelerated perpendicular to the wire, so energy is not absorbed from the perpendicular component of the electric field.

If we have a grid of parallel wires, and an electromagnetic wave is incident on it, only the component of the electric field transverse to the grid will pass through. Thus the transmitted wave is linearly polarized, in the direction transverse to that of the grid. We can call this the **axis** of the polarizing filter.

Edwin Land's polarizing filter works in exactly the same way, but the "wires" are in fact long molecules.

It can be shown that the intensity of an electromagnetic wave is proportional to the amplitude squared of the electric field. So if a linearly polarized wave is incident on a polarizing filter with intensity I_0,

$$I_0 \propto E_0^2 \tag{9.2}$$

Once it passes through the filter, the component of the electric field in

the direction of the filter's axis is removed. So if θ is the angle between $\vec{E_0}$ and the axis of the filter, the intensity I after the filter is

$$I \propto (E_0 \cos\theta)^2 \tag{9.3}$$

Therefore,

$$I = I_0 \cos^2\theta \tag{9.4}$$

This result is known as **Malus' law**. If the electric field is aligned with the axis of the filter, $\theta = 0$ and $I = I_0$; the wave passes through the filter without any loss. On the other hand, if $\vec{E_0}$ is perpendicular to the axis of the filter, $\theta = \pi/2$ and $I = 0$; the wave is completely extinguished by the filter.

By measuring I and I_0, it is possible to deduce the angle θ, i.e. the orientation of $\vec{E}$ of the original wave. When a polarizing filter is used in this way, it is also known as a **analyzer**.

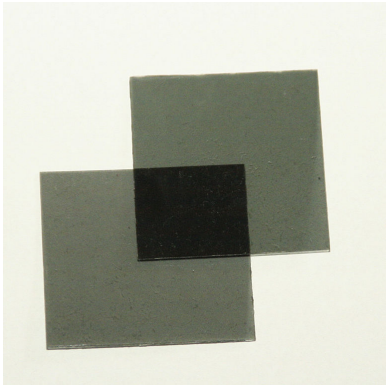


Figure 9.3: Where two crossed polarizers overlap, no light is transmitted. Notice also that the light transmitted by either one of the polarizers alone is reduced in intensity. The intensity of natural unpolarized light is reduced by half, as discussed in the text below. Credit: Niels Bosboom.

9.4 *The effect of a polarizing filter on natural unpolarized light*

Unpolarized light from a source like an incandescent light bulb, or the Sun[1] may be envisioned as wave trains emitted with random orientations of the electric field. If such a beam passes through a polarizing filter, each wave train will be reduced in intensity to a greater or lesser degree, depending on the orientation of its electric field relative to the axis of the filter. If we let $\langle\ \rangle$ denote an average, the average intensity of the waves after passing through the filter is given by averaging equation (9.4):

$$\langle I \rangle = \langle I_0 \rangle \langle \cos^2\theta \rangle \tag{9.5}$$

where θ varies randomly between 0 and π. It's not hard to prove that $\langle \cos^2\theta \rangle = 1/2$, and therefore

$$\langle I \rangle = \frac{1}{2}\langle I_0 \rangle \tag{9.6}$$

So by letting unpolarized light pass through a polarizing filter, its intensity is reduced to one half of the original intensity[2]. And of course, it also emerges linearly polarized in the direction of the axis of the filter, since for each wave train the component of the electric field perpendicular to the filter axis is extinguished.

So a polarizing filter can be used to produce linearly polarized light from unpolarized light. When used in this way, it is also called a **polarizer**.

[1] Although the sky is actually slightly polarized; but we will ignore that for the moment.

[2] This is why polarized sunglasses look dark.

9.5 *Polarization by reflection*

There is another way of producing linearly polarized light, that doesn't involve filters at all. When light is incident on an interface between

two transparent media of different refractive indices, part of is reflected and part of it is transmitted. As with acoustic waves, we can calculate a reflection coefficient $\mathcal{R}$, which is the ratio between the reflected amplitude and the incident amplitude, which depends on the ratio between the refractive indices of the two media, just as $\mathcal{R}$ for acoustic waves depended on the acoustic impedance ratio.

Previously, when we discussed acoustic waves, we only worked with waves at **normal incidence**, that is at right angles to the interface. But of course it is possible to have a wave incident at any angle θ, and we might expect $\mathcal{R}$ to be a function of θ, and that is indeed what happens.

So with electromagnetic waves also, $\mathcal{R}$ depends on the ratio of the refractive indices, and the angle of incidence. But in the electromagnetic case, it also depends on the polarization of the incoming wave[3]. We distinguish between two cases:

[3] Acoustic waves are longitudinal, so there is no polarization.

1. The electric field vector oscillates in the plane perpendicular to the interface, and it is said to be s-polarized (from the German *senkrecht*, meaning perpendicular); and

2. the electric field vector oscillates in the plane parallel to the interface, and it is said to be p-polarized.

Another terminology often used is "transverse electric" (TE) for s-polarization, and "transverse magnetic" (TM) for p-polarization (because when the electric field vector is parallel to the interface, the magnetic field vector is transverse to it).

Figure 9.4 below shows plots of $\mathcal{R}$ vs. incidence angle θ for light going between air ($n = 1$) and glass ($n = 1.5$). In each case, the light is passing from medium 1 into medium 2.

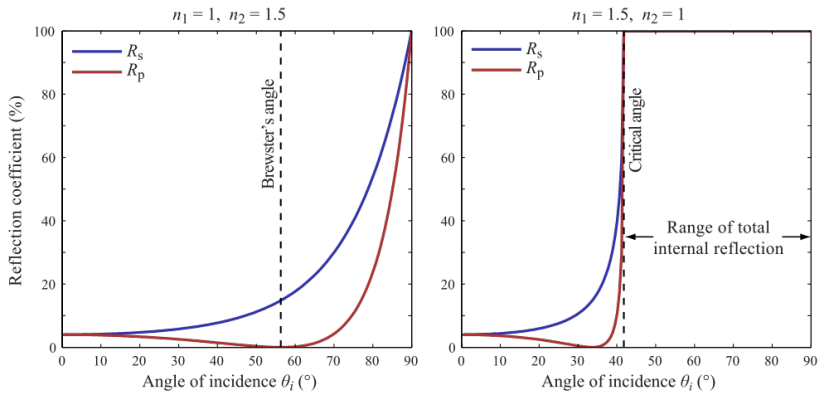


Figure 9.4: Reflection coefficients vs. incidence angle θ for light passing between air and glass.

In both cases $\mathcal{R}_s \neq \mathcal{R}_p$, as discussed above. But there is also something particularly interesting about the TM reflection coefficient $\mathcal{R}_p$: At a particular angle, $\mathcal{R}_p = 0$. In the case of the light going from air into glass, it is called **Brewster's angle**. Brewster's angle also depends on the ratio between the refractive indices: It can be shown that

$$\tan\theta_B = \frac{n_2}{n_1} \tag{9.7}$$

For example, in the case of Figure 9.4, $\theta_B = \arctan(1.5) = 56.3°$.

This means that whatever the polarization is of the incoming light, at that particular angle of incidence the component that is polarized parallel to the interface is not reflected, so the reflected light is plane polarized, in the direction perpendicular to the interface. So we have a way of producing linearly polarized light, simply by bouncing light off an interface between two media, at Brewster's angle, or close to it.

This is also a good way of getting rid of unwanted reflections at angles around Brewster's angle: intercept the light with a polarizer, with its axis parallel to the interface. This is the idea behind polarized sunglasses, which reduce the glare from the road surface, particularly when it is wet.

Figure 9.5 shows the sky reflected in a puddle after a rain shower. Figure 9.6 shows the same puddle, now through a polarizing filter with its axis horizontal. The reflection has all but disappeared.

And before we leave the topic of reflection at an interface altogether, notice also that in the case of light going from glass into air, there is no transmitted light at angles greater than the critical angle.



Figure 9.5: The sky reflected in a puddle. No filter used.



Figure 9.6: The same scene, now photographed through a polarizer, with its axis parallel to the surface.

9.6 *The strange case of the birefringent crystal*

Figure 9.7 show a pattern seen through a **calcite** crystal. The crystal refracts light like any other transparent material, but it has also the peculiar property that it produces not one but two different refracted rays! This property is known as **birefringence**.



Figure 9.7: Birefringence in a calcite crystal

The explanation of this strange behaviour is that the material actually has two different refractive indices, depending on the orientation of the electric field vector. One can define an **optical axis** in the crystal, so that if the incident electric field vector is seen as the sum of two components, one parallel and the other perpendicular to the optical axis, the corresponding waves will travel at different speeds inside the crystal. (And therefore, by Fermat's principle, follow different paths inside the crystal). Light whose $\vec{E}$ vector is *perpendicular* to the optical axis is called the **ordinary ray**, and light whose $\vec{E}$ vector is *parallel* to the optical axis is called the **extraordinary ray**.

It follows that each outgoing ray is plane polarized, and at right angles to each other. This can be easily verified by looking at the refracted rays through a polarizing filter.

To visualize this more clearly, let us draw the electric field vector once again in the (x, y) plane, perpendicular to the propagation direction, which will be the z axis, like we did in Figure 9.2. *But now, let the y axis represent the optical axis of the crystal.* This means that the incidence angle is zero, and therefore the angle of refraction is also zero. So the ordinary and extraordinary rays follow the same path in this case, both

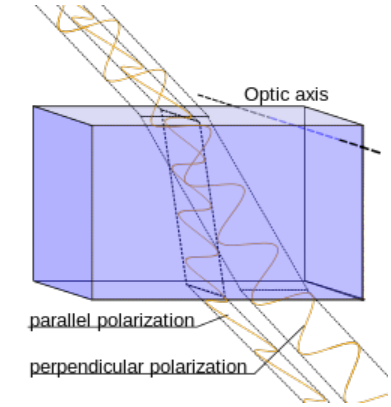


Figure 9.8: In a birefringent material, rays parallel to the optical axis are refracted differently from rays perpendicular to it.

propagating along the z axis, but at different speeds.

Let's say that we have plane polarized light incident on the crystal at $z = 0$, and for simplicity we will make $E_{0x} = E_{0y} = E_0$[4]. As the wave travels through the birefringent medium, both components have the same frequency, but the x component has a wave number k_O such that $ck_O/\omega = n_O$ (remember that the x axis is the ordinary axis), whereas the y component has a different wave number k_E such that $ck_E/\omega = n_E$:

[4] That means the electric field of the incident wave arrives at 45° from the x axis.

$$\tilde{E}_x = E_0 e^{i(\omega t - k_O z)} \tag{9.8a}$$

$$\tilde{E}_y = E_0 e^{i(\omega t - k_E z)} \tag{9.8b}$$

After advancing through the thickness d of the crystal, the waves emerge out of phase with each other:

$$\tilde{E}_x = E_0 e^{i(\omega t - k_O d)} \tag{9.9a}$$

$$\tilde{E}_y = E_0 e^{i(\omega t - k_E d)} = E_0 e^{i(\omega t - k_O d - \phi)} = \tilde{E}_x e^{-i\phi} \tag{9.9b}$$

where

$$\phi = (k_E - k_O)d = \frac{\omega d}{c}(n_E - n_O) \tag{9.10}$$

Furthermore, since c/ω is the wave number k of the incident wave, and $k = 2\pi/\lambda$, where λ is the wavelength of the incident wave, we can write this expression for ϕ as

$$\phi = \frac{2\pi}{\lambda} d(n_E - n_O) \tag{9.11}$$

There are several possible outcomes, depending on the value of ϕ. (Take a look at the simulation at http://www.geogebratube.org/material/show/id/297907).

- If $\phi = 2\pi$, the outgoing wave is linearly polarized, at 45° from the x axis, exactly as before entering the birefringent medium;
- if $\phi = \pi$, the outgoing wave is again linearly polarized, but at 135° from the x axis;
- if $\phi = \pi/2$, the electric field vector traces out a circle, in the counterclockwise, or left-hand, direction. The light is said to be **left circular polarized**;
- if $\phi = 3\pi/2$, the electric field vector traces out a circle again, but in the clockwise, or right-hand, direction. The light is said to be **right circular polarized**;
- for all other values of ϕ, the electric field vector traces out an ellipse. The light is said to be **elliptically polarized**.

In particular, let's look at the conditions needed to produce left circular polarization. The phase difference between the ordinary and extraordinary rays must be $\phi = \pi/2$, or in general $\phi = (2m-1)\pi/2$, where $m = 0, 1, 2, \ldots$, and ϕ is given by equation (9.11). So, given a particular material with particular values of n_E and n_O, ϕ depends on the thickness of the medium, d. So we can turn the condition for ϕ into a condition for d:

$$d = \frac{2m-1}{n_E - n_O}\frac{\lambda}{4} \tag{9.12}$$

This is known as a **quarter-wave plate**. It is used to produce circular polarized from linearly polarized light.